

CaF–He Collision Estimate: Formulas and Distance Tables

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Estimate of the number of CaF–He collisions from the buffer-gas-cell nozzle exit ($z = 0$) out to $z = 1$ m, for a cryogenic buffer-gas source adapted for pulsed-ablation CaF production on the continuous-flow cell design of Gangwar *et al.*, “Buffer gas cooling of a continuous CO molecular beam,” *Optics Express* **33**, 28164 (2025), doi:10.1364/OE.560951.

Input parameters

Quantity	Symbol	Value	Source
Cell / nozzle temperature	T	4 K	user-specified (cell 4 K stage; 3.7 K measured in the paper)
He flow rate	f_{He}	20 sccm $= 9.0\text{e}18\text{ s}^{-1}$	user-specified; 1 sccm = $4.5\text{e}17\text{ s}^{-1}$ (paper’s own conversion)
Nozzle diameter	d	3 mm and 5 mm	range tested in the paper (2–5 mm); brackets typical CaF cells
CaF–He cross section	σ	$1\text{e}-14\text{ cm}^2$	order-of-magnitude value used throughout buffer-gas literature (paper uses the same value for CO–He)
Background chamber pressure	P_{bg}	$1\text{e}-7$ mbar	paper’s stated typical operating pressure, both chambers
Background gas temperature	T_{bg}	300 K	gauge reads pressure referenced to room temperature

Two collision zones are combined: (1) a *near-field* zone right at the nozzle where the He is still dense and the flow is expanding hydrodynamically (this is where essentially all the collisions happen), and (2) a *far-field* zone where the molecule coasts through the low residual chamber pressure. Collisions inside the buffer-gas cell itself (thermalization, upstream of $z = 0$) are not included here.

Formulas used

1. Mean thermal speed of He at temperature T

$$\bar{v}_{He} = \sqrt{\frac{8k_B T}{\pi m_{He}}} \quad (1)$$

2. He number density at the aperture (stagnation density)

$$n_0 = \frac{4f_{He}}{A_{aperture} \bar{v}_{He}}, \quad A_{aperture} = \frac{\pi d^2}{4} \quad (2)$$

where f_{He} is the He flow in particles/s (1 sccm = $4.5\text{e}17\text{ s}^{-1}$).

3. He–He mean free path near the aperture

$$\lambda_{He} = \frac{1}{\sqrt{2} n_0 \sigma_{He-He}} \quad (3)$$

4. Reynolds number of the expansion

$$Re \approx \frac{2d}{\lambda_{He}} \quad (4)$$

5. Total collisions picked up near the orifice

$$N_{\text{near-field}} \approx Re/2 \quad (5)$$

(standard buffer-gas-beam result, Hutzler *et al.*, *Chem. Rev.* **112**, 4803 (2012); used directly in the source paper.)

6. Local He density vs. distance downstream (free-jet scaling)

$$n(z) = \begin{cases} n_0, & z \leq z^* \\ 0.15 n_0 \left(\frac{d}{z}\right)^2, & z > z^* \end{cases} \quad \text{floored at } n_{bg} \text{ if smaller,} \quad z^* = \sqrt{0.15} d \quad (6)$$

7. Local collision rate (inverse local mean free path)

$$\frac{dN}{dz}(z) = n(z) \sigma_{CaF-He} \quad [\text{cm}^{-1}] \quad (7)$$

8. Cumulative collisions from nozzle exit to distance z

$$N(z) = \int_0^z \frac{dN}{dz'} dz' = N_{\text{near-field}} \cdot f(z) + n_{bg} \sigma_{CaF-He} z \quad (8)$$

$$f(z) = \frac{z}{2z^*} \quad (z \leq z^*), \quad f(z) = 1 - \frac{z^*}{2z} \quad (z > z^*) \quad (9)$$

where $n_{bg} = P_{bg}/(k_B T_{bg})$ is the background chamber density from the gauge reading.

9. Background (far-field) contribution

$$N_{bg}(z) = n_{bg} \sigma_{CaF-He} z \quad (10)$$

Note: the near-field density profile $n(z)$ (item 6) is a standard free-jet-expansion scaling (z^{-2} falloff past a few nozzle diameters), not itself taken from the paper; it is normalized so that its integral reproduces the paper's cited near-orifice total, $N_{\text{near-field}} \approx Re/2$. The z^{-2} shape and the $Re/2$ total are therefore on different footings: the total is a literature-standard result, the distance-dependence is a reasonable modeling choice calibrated to it.

Derived quantities for the two nozzle sizes

Quantity	3 mm nozzle	5 mm nozzle
Aperture area A (cm ²)	0.07069	0.19635
Cell/aperture density n_0 (cm ⁻³)	3.501e16	1.2605e16
He-He mean free path (μ m)	20.20	56.10
Reynolds number Re	297.1	178.3
Near-field total, $Re/2$	148.5	89.13
Matching distance z^* (mm)	1.162	1.936
Background density n_{bg} (cm ⁻³)	2.414e9	2.414e9

Results: 3 mm nozzle

Fine-resolution table, 0–10 mm (0.5 mm steps)

This is where essentially all the near-field collisions occur.

z (mm)	rate dN/dz (cm^{-1})	collisions in 0.5-mm bin	cumulative $N(z)$
0	350.126	0	0
0.5	350.126	31.962	31.962
1	350.126	31.962	63.9239
1.5	210.076	27.0903	91.0143
2	118.167	14.3829	105.397
2.5	75.6272	8.62973	114.027
3	52.5189	5.75316	119.78
3.5	38.5853	4.1094	123.889
4	29.5419	3.08205	126.971
4.5	23.3417	2.39715	129.369
5	18.9068	1.91772	131.286
5.5	15.6255	1.56904	132.855
6	13.1297	1.30754	134.163
6.5	11.1875	1.10638	135.269
7	9.64632	0.948323	136.218
7.5	8.40302	0.82188	137.04
8	7.38547	0.719146	137.759
8.5	6.54214	0.63454	138.393
9	5.83543	0.564036	138.957
9.5	5.23734	0.504664	139.462
10	4.7267	0.454198	139.916

Per-centimeter table, 0–100 cm (1 cm steps)

Rate and per-bin collisions collapse to the residual background-gas value beyond a few cm; cumulative total asymptotes to $Re/2 = 148.5$.

z (cm)	rate dN/dz (cm^{-1})	collisions in 1-cm bin	cumulative $N(z)$
0	350.126	0	0
1	4.7267	139.916	139.916
2	1.18167	4.31489	144.231
3	0.525189	1.43831	145.669
4	0.295419	0.719168	146.388
5	0.189068	0.431511	146.82
6	0.131297	0.287682	147.108
7	0.0964632	0.205494	147.313
8	0.0738547	0.154127	147.467
9	0.0583543	0.119882	147.587
10	0.047267	0.0959101	147.683

z (cm)	rate dN/dz (cm ⁻¹)	collisions in 1-cm bin	cumulative $N(z)$
11	0.0390636	0.0784763	147.762
12	0.0328243	0.0654009	147.827
13	0.0279686	0.0553429	147.882
14	0.0241158	0.0474403	147.93
15	0.0210076	0.0411181	147.971
16	0.0184637	0.0359814	148.007
17	0.0163554	0.0317511	148.039
18	0.0145886	0.0282259	148.067
19	0.0130933	0.0252573	148.092
20	0.0118167	0.022734	148.115
21	0.0107181	0.0205711	148.135
22	0.00976591	0.0187032	148.154
23	0.00893516	0.0170789	148.171
24	0.00820607	0.0156577	148.187
25	0.00756272	0.014407	148.201
26	0.00699216	0.0133007	148.215
27	0.00648381	0.0123172	148.227
28	0.00602895	0.0114391	148.238
29	0.00562033	0.0106519	148.249
30	0.00525189	0.00994338	148.259
31	0.00491852	0.00930342	148.268
32	0.00461592	0.00872347	148.277
33	0.0043404	0.00819624	148.285
34	0.00408884	0.00771553	148.293
35	0.00385853	0.00727602	148.3
36	0.00364714	0.00687314	148.307
37	0.00345267	0.00650292	148.313
38	0.00327334	0.00616193	148.32
39	0.00310763	0.00584717	148.325
40	0.00295419	0.00555602	148.331
41	0.00281184	0.00528617	148.336
42	0.00267953	0.0050356	148.341
43	0.00255635	0.00480251	148.346
44	0.00244148	0.00458531	148.351
45	0.00233417	0.00438259	148.355
46	0.00223379	0.0041931	148.359
47	0.00213975	0.00401569	148.363
48	0.00205152	0.00384938	148.367
49	0.00196864	0.00369325	148.371
50	0.00189068	0.00354648	148.374
51	0.00181726	0.00340835	148.378
52	0.00174804	0.00327819	148.381
53	0.0016827	0.0031554	148.384
54	0.00162095	0.00303942	148.387
55	0.00156255	0.00292978	148.39
56	0.00150724	0.002826	148.393

z (cm)	rate dN/dz (cm ⁻¹)	collisions in 1-cm bin	cumulative $N(z)$
57	0.00145482	0.00272769	148.396
58	0.00140508	0.00263447	148.398
59	0.00135786	0.00254598	148.401
60	0.00131297	0.00246192	148.403
61	0.00127028	0.00238199	148.406
62	0.00122963	0.00230593	148.408
63	0.0011909	0.0022335	148.41
64	0.00115398	0.00216445	148.413
65	0.00111875	0.0020986	148.415
66	0.0010851	0.00203574	148.417
67	0.00105295	0.00197569	148.419
68	0.00102221	0.00191829	148.421
69	0.000992795	0.00186339	148.422
70	0.000964632	0.00181084	148.424
71	0.000937651	0.00176051	148.426
72	0.000911786	0.00171228	148.428
73	0.000886977	0.00166603	148.429
74	0.000863166	0.00162165	148.431
75	0.000840302	0.00157905	148.433
76	0.000818334	0.00153813	148.434
77	0.000797217	0.00149881	148.436
78	0.000776906	0.001461	148.437
79	0.000757362	0.00142462	148.438
80	0.000738547	0.00138961	148.44
81	0.000720424	0.00135589	148.441
82	0.000702959	0.00132341	148.443
83	0.000686123	0.0012921	148.444
84	0.000669884	0.00126191	148.445
85	0.000654214	0.00123279	148.446
86	0.000639089	0.00120468	148.448
87	0.000624481	0.00117754	148.449
88	0.000610369	0.00115133	148.45
89	0.00059673	0.001126	148.451
90	0.000583543	0.00110151	148.452
91	0.000570788	0.00107783	148.453
92	0.000558447	0.00105493	148.454
93	0.000546502	0.00103276	148.455
94	0.000534936	0.0010113	148.456
95	0.000523734	0.000990519	148.457
96	0.00051288	0.000970386	148.458
97	0.000502359	0.000950876	148.459
98	0.000492159	0.000931963	148.46
99	0.000482267	0.000913623	148.461
100	0.00047267	0.000895833	148.462

Results: 5 mm nozzle

Fine-resolution table, 0–10 mm (0.5 mm steps)

This is where essentially all the near-field collisions occur.

z (mm)	rate dN/dz (cm^{-1})	collisions in 0.5-mm bin	cumulative $N(z)$
0	126.045	0	0
0.5	126.045	11.5063	11.5063
1	126.045	11.5063	23.0126
1.5	126.045	11.5063	34.5189
2	118.167	11.4599	45.9788
2.5	75.6272	8.62973	54.6086
3	52.5189	5.75316	60.3617
3.5	38.5853	4.1094	64.4711
4	29.5419	3.08205	67.5532
4.5	23.3417	2.39715	69.9503
5	18.9068	1.91772	71.868
5.5	15.6255	1.56904	73.4371
6	13.1297	1.30754	74.7446
6.5	11.1875	1.10638	75.851
7	9.64632	0.948323	76.7993
7.5	8.40302	0.82188	77.6212
8	7.38547	0.719146	78.3403
8.5	6.54214	0.63454	78.9749
9	5.83543	0.564036	79.5389
9.5	5.23734	0.504664	80.0436
10	4.7267	0.454198	80.4978

Per-centimeter table, 0–100 cm (1 cm steps)

Rate and per-bin collisions collapse to the residual background-gas value beyond a few cm; cumulative total asymptotes to $Re/2 = 89.13$.

z (cm)	rate dN/dz (cm^{-1})	collisions in 1-cm bin	cumulative $N(z)$
0	126.045	0	0
1	4.7267	80.4978	80.4978
2	1.18167	4.31489	84.8127
3	0.525189	1.43831	86.251
4	0.295419	0.719168	86.9702
5	0.189068	0.431511	87.4017
6	0.131297	0.287682	87.6893
7	0.0964632	0.205494	87.8948
8	0.0738547	0.154127	88.049
9	0.0583543	0.119882	88.1688
10	0.047267	0.0959101	88.2648

z (cm)	rate dN/dz (cm ⁻¹)	collisions in 1-cm bin	cumulative $N(z)$
11	0.0390636	0.0784763	88.3432
12	0.0328243	0.0654009	88.4086
13	0.0279686	0.0553429	88.464
14	0.0241158	0.0474403	88.5114
15	0.0210076	0.0411181	88.5525
16	0.0184637	0.0359814	88.5885
17	0.0163554	0.0317511	88.6203
18	0.0145886	0.0282259	88.6485
19	0.0130933	0.0252573	88.6738
20	0.0118167	0.022734	88.6965
21	0.0107181	0.0205711	88.7171
22	0.00976591	0.0187032	88.7358
23	0.00893516	0.0170789	88.7528
24	0.00820607	0.0156577	88.7685
25	0.00756272	0.014407	88.7829
26	0.00699216	0.0133007	88.7962
27	0.00648381	0.0123172	88.8085
28	0.00602895	0.0114391	88.82
29	0.00562033	0.0106519	88.8306
30	0.00525189	0.00994338	88.8406
31	0.00491852	0.00930342	88.8499
32	0.00461592	0.00872347	88.8586
33	0.0043404	0.00819624	88.8668
34	0.00408884	0.00771553	88.8745
35	0.00385853	0.00727602	88.8818
36	0.00364714	0.00687314	88.8886
37	0.00345267	0.00650292	88.8951
38	0.00327334	0.00616193	88.9013
39	0.00310763	0.00584717	88.9072
40	0.00295419	0.00555602	88.9127
41	0.00281184	0.00528617	88.918
42	0.00267953	0.0050356	88.923
43	0.00255635	0.00480251	88.9278
44	0.00244148	0.00458531	88.9324
45	0.00233417	0.00438259	88.9368
46	0.00223379	0.0041931	88.941
47	0.00213975	0.00401569	88.945
48	0.00205152	0.00384938	88.9489
49	0.00196864	0.00369325	88.9526
50	0.00189068	0.00354648	88.9561
51	0.00181726	0.00340835	88.9595
52	0.00174804	0.00327819	88.9628
53	0.0016827	0.0031554	88.9659
54	0.00162095	0.00303942	88.969
55	0.00156255	0.00292978	88.9719
56	0.00150724	0.002826	88.9747

z (cm)	rate dN/dz (cm ⁻¹)	collisions in 1-cm bin	cumulative $N(z)$
57	0.00145482	0.00272769	88.9775
58	0.00140508	0.00263447	88.9801
59	0.00135786	0.00254598	88.9826
60	0.00131297	0.00246192	88.9851
61	0.00127028	0.00238199	88.9875
62	0.00122963	0.00230593	88.9898
63	0.0011909	0.0022335	88.992
64	0.00115398	0.00216445	88.9942
65	0.00111875	0.0020986	88.9963
66	0.0010851	0.00203574	88.9983
67	0.00105295	0.00197569	89.0003
68	0.00102221	0.00191829	89.0022
69	0.000992795	0.00186339	89.0041
70	0.000964632	0.00181084	89.0059
71	0.000937651	0.00176051	89.0077
72	0.000911786	0.00171228	89.0094
73	0.000886977	0.00166603	89.011
74	0.000863166	0.00162165	89.0127
75	0.000840302	0.00157905	89.0142
76	0.000818334	0.00153813	89.0158
77	0.000797217	0.00149881	89.0173
78	0.000776906	0.001461	89.0187
79	0.000757362	0.00142462	89.0202
80	0.000738547	0.00138961	89.0215
81	0.000720424	0.00135589	89.0229
82	0.000702959	0.00132341	89.0242
83	0.000686123	0.0012921	89.0255
84	0.000669884	0.00126191	89.0268
85	0.000654214	0.00123279	89.028
86	0.000639089	0.00120468	89.0292
87	0.000624481	0.00117754	89.0304
88	0.000610369	0.00115133	89.0315
89	0.00059673	0.001126	89.0327
90	0.000583543	0.00110151	89.0338
91	0.000570788	0.00107783	89.0349
92	0.000558447	0.00105493	89.0359
93	0.000546502	0.00103276	89.0369
94	0.000534936	0.0010113	89.038
95	0.000523734	0.000990519	89.0389
96	0.00051288	0.000970386	89.0399
97	0.000502359	0.000950876	89.0409
98	0.000492159	0.000931963	89.0418
99	0.000482267	0.000913623	89.0427
100	0.00047267	0.000895833	89.0436

Caveats

$\sigma_{CaF-He} = 1\text{e-}14\text{ cm}^2$ is a generic order-of-magnitude value for molecule–He pairs used across the buffer-gas-beam literature; the true CaF–He elastic cross section has energy-dependent structure (see e.g. Londoño & Pérez-Ríos, *J. Chem. Phys.* **162**, 234304 (2025)) and could differ by a factor of a few. The near-field z -dependence (item 6 in the formulas) is a standard free-jet scaling normalized to reproduce the paper’s own $Re/2$ total, not a first-principles result for this specific cell. Background pressure is taken as the paper’s own quoted steady-state operating value ($1\text{e-}7$ mbar); the transient gas/plasma load from the ablation pulse itself is not included.